Pre-Performance Routines among Club Volleyball Players: The Relationship between Routines and Accuracy in Serving

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INTRODUCTION

- Routines help prepare athletes to complete a set of automatic skills despite variations in the sporting context (McCann, 2008; Taylor, 2012; Wrisberg & Pein, 1992).
- Many athletes use pre-performance routines (PPR) during competition to help improve performance (Bell, Cox, & Finch, 2010; Boutcher & Zinsser, 1990; Thomas & Over, 1994).
- PPRs involve an athlete focusing solely on a specific sequence of pre-established and rehearsed cues prior to a sport specific, closed skill (e.g. a punt in football or soccer, batting stance in baseball; Wrisberg & Pein, 1992).

The current study on PPRs extends previous research by examining volleyball and including a larger sample size, both genders, various ages and experience levels, and by examining PPRs in competition rather than scrimmages or practices (Bell, Cox, & Finch, 2010).

- **Primary aim** was to examine the relationship between the use of a behavioral PPR and serving accuracy among collegiate club volleyball players. It was hypothesized that maintainers of PPRs would exhibit higher serving accuracy than non-maintainers.
- **Secondary aim** was to examine the relationship between PPRs and competitive anxiety (CA) with the hypothesis that maintainers would exhibit lower levels of CA.
- **Tertiary aim** was to examine potential gender differences with the hypothesis that females would be more likely to use PPRs than males.

METHODS

- 40 (21 females and 19 males) intercollegiate club volleyball players
- Participants completed a demographic form, Competitive State Anxiety Index-2(d) (CSAI-2(d)) and a modified version designed to look at serving. Then video recordings were taken of each participant's serve.
- PI and research staff then recorded participants' serving at six different regional intercollegiate competitions in order to observe each team twice. The current score and the result of the serve (i.e. "make" or "miss") was also verbally recorded.
- Participants were coded either as maintainers (maintained actions of PPR ≥90%; MTR) or non-maintainers (maintained actions of PPR 0-30%; NMTR).
- Independent t-tests were used to compare differences between MTRs and NMTRs on serving percentage, scores on the CSAI-2(d) and the modified version, and gender differences in years and level of experience and serving percentage.







RESULTS

Primary Aim: There was no difference between MTR and NMTR on serving accuracy (t= 1.245; *p*= .250). Below upper left: Mean demographic data of behavioral and temporal serve PPRs and serve percentage.

Secondary Aim: There were no significant differences between the MTR and NMTR groups or men and women on the any of the subscales in either version. Below right: CASI-2(d) and modified sub-scale scores.

Tertiary Aim: Overall, there were no gender differences on any of the general or volleyball specific demographics Women did, however, maintain their behavioral serve PPR (72.05%) at a significantly higher percentage than the men (44.63%), *t*= 3.54, *p*= .001. Below lower left: Mean demographic data by gender.

Group	Number of Serves Made	Number of Serves Attempted	Serving %	Number of Participants	Average Maintenance %
Behavioral Maintenance					
MTR	195	213	90.53 %	4 (4 women, 0 men)	97.54%
NMTR	129	147	85.74%	5 (1 women, 4 men)	29.26%
Remaining	506	580	89.36%	20 (9 women, 11 men)	55.78%

Gender	Age	Classification	Ethnicity	Years of Experience	Experience Level
Women M	19.76	Sophomore	90.47% Caucasian	8-12	Advanced
Men M	19.42	Freshman	84.21% Caucasian	4-8	Advanced

Sub-Scales	Mean	Directional		
		Minimum-Maximum	Mean	Minimum-Maximum
CSAI-2(d)				
Cognitive Anxiety				
Women	20.39	9 – 31	.39	-12 - +27
Men	21.58	14 – 28	3.32	-11 - +19
Total	20.90	9 - 31	1.925	-12 - +27
Somatic Anxiety				
Women	17.27	9 - 25	4.62	-9 - +27
Men	19.08	11 - 28	4.42	-7 - +19
Total	18.05	9 - 28	7.739	-9 - +27
Self-Confidence				
Women	24.79	12 - 36	9.54	-10 - +27
Men	26.78	23 - 33	14.11	-3 - +21
Total	25.87	12 - 36	9.076	-10 - +27
CSAI-2(d) Modified				
Cognitive Anxiety				
Women	20.81	9 - 30	-0.61	-19 - +27
Men	20.73	13 - 28	-0.44	-19 - +20
Total	20.794	9 - 30	.307	-19 - +27
Somatic Anxiety				
Women	16.18	9 - 29	3.92	-18 - +27
Men	16.48	9 - 27	2.5	-18 - +23
Total	16.153	9 - 29	4.384	-18 - +27
Self-Confidence				
Women	25.22	16 - 36	9.38	-18 - +27
Men	26.09	16 - 36	12.8	-9 - +27
Total	25.692	16 - 36	11.205	-18 - +27

CONCLUSIONS

Primary aim showed no significant differences between the MTR and NMTR groups on serving accuracy.

- This finding did not support the hypothesis, but did confirm similar findings by Czech, Ploszay, and Burke (2004).
- In volleyball strict serve PPR maintenance may not be as important for successful performance because players are navigating spectator movement, variations in whistle speed, and changes in space for serving.

Secondary aim exploring CA, found a greater reported amount of cognitive compared to somatic anxiety. May be because participants were all relatively advanced players.

- Subscale of self-confidence warrants the most attention for performance (Craft, Magyar, Becker, & Feltz, 2003).
- Feelings of self-confidence may mitigate the anxiety and allow the participants to attribute it more positively.

Tertiary aim found no general or volleyball specific demographic differences between the two genders, although women reported a greater number of years of experience.

- Findings were consistent with the hypothesis that women were significantly higher maintainers than their male counterparts.
- However, this did not correlate to their serving percentage.

Future research should seek to develop study designs that use deception to get a more natural depiction of a PPR, assess cognitive components of PPRs, and more rigorously define accuracy as it suits volleyball in particular.

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